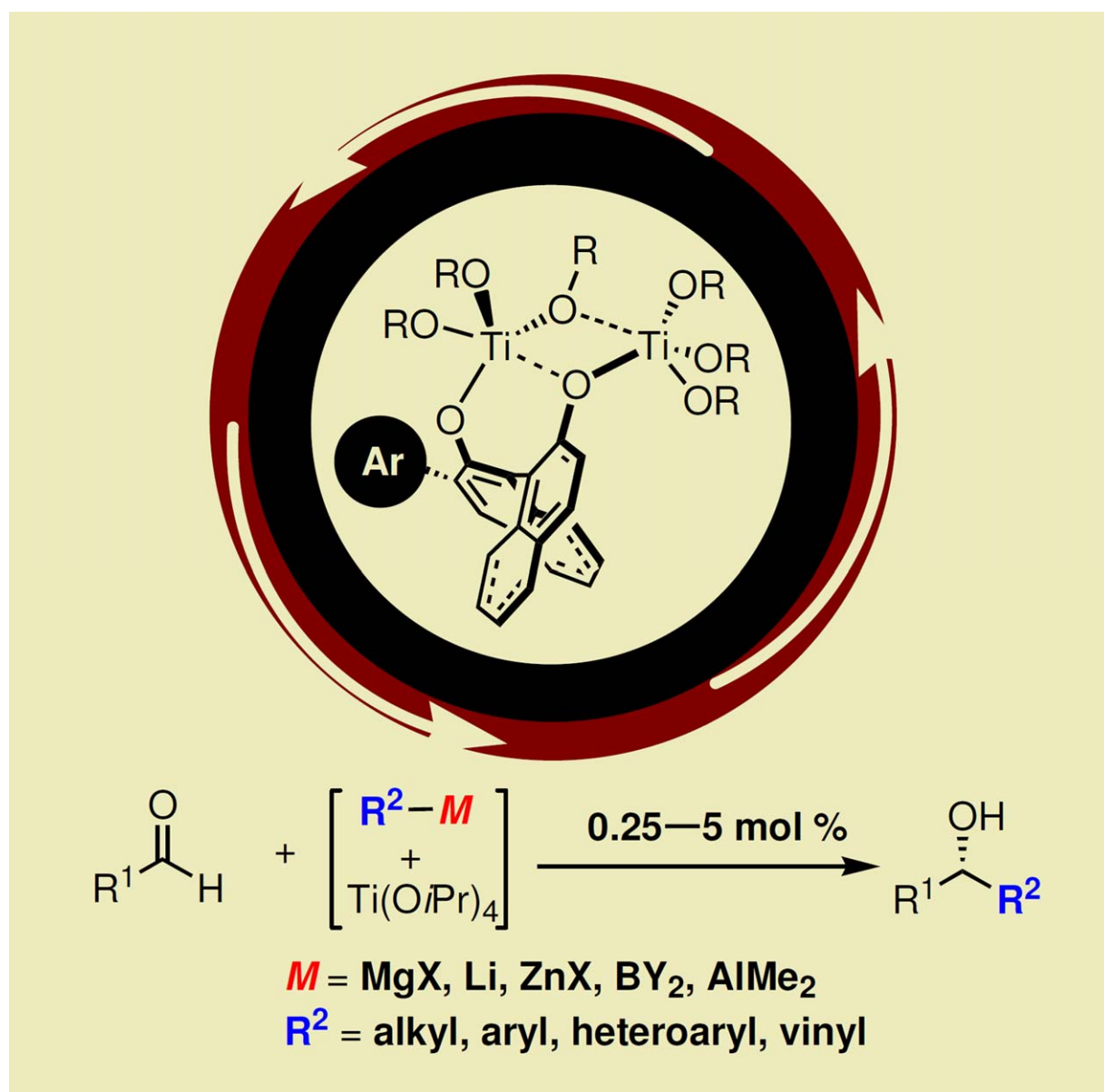


Development of Highly Active Chiral Titanium Catalysts for the Enantioselective Addition of Various Organometallic Reagents to Aldehydes

Toshiro Harada



ABSTRACT: The catalytic enantioselective carbonyl addition reaction has long attracted the interest of chemists because of its synthetic importance. Although many highly enantioselective reactions have been developed, with few exceptions the reactions are carried out at relatively high catalyst loadings, making them less practical for scale-up applications. In addition, organometallic reagents employed as carbon nucleophiles have been limited to those with relatively low reactivity, such as diorganozincs and arylboronic acids. In an effort to enhance the practicality, a highly active and enantioselective chiral titanium catalyst system was recently developed in our laboratory, enabling the enantioselective carbonyl addition reaction to aldehydes using various organometallic reagents (RM; M = MgX, Li, BY₂, ZnX, AlMe₂) at lower catalyst loadings (≤ 5 mol %).

Keywords: alcohols, asymmetric catalysis, asymmetric synthesis, nucleophilic addition, titanium

1. Introduction

1.1. Ligand Acceleration

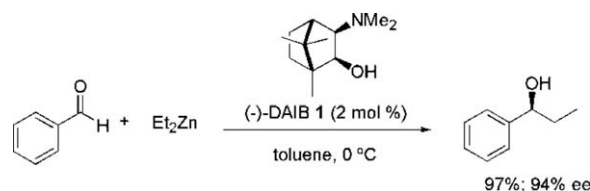
The addition of organometallic reagents to carbon–heteroatom double bonds in aldehydes, ketones, imines, and so on is a fundamental reaction in organic synthesis, providing alcohols and amines with simultaneous carbon–carbon bond formation. Since the early days, there has been a continuing interest in synthesizing one of the enantiomers selectively through these reactions. The first attempt was reported in 1940 by Betti and Lucchi, who examined the Grignard reaction in the presence of (+)-bornyldimethylamine.^[1] Although the results could not be reproduced later, Cohen and Wright reported an appreciable selectivity in the Grignard reaction with (2*R*,3*R*)-2,3-dimethoxybutane in 1953.^[2] After the late 1960s, many chemists joined in this research area and made their contributions. Enantioselectivities higher than 95% *ee* were reported in the early 1980s.^[3,4] However, all reactions required more than a stoichiometric amount of chiral additives or modifiers. The catalytic enantioselective carbonyl addition appeared to be far away from realization at that time.

In the mid-1980s, a breakthrough was made with the discovery of a chiral amino alcohol catalyzed enantioselective addition of diethylzinc to aldehydes by Oguni et al.^[5] and by the subsequent development of the highly efficient DAIB catalyst **1** by Noyori and co-workers. (Scheme 1).^[6] Only 2 mol % of the amino alcohol is sufficient to achieve high enantioselectivity. In comparison with previously employed Grignard and organolithium reagents, dialkylzincs are much less reactive. However, once modified by the appropriate chiral amino alco-

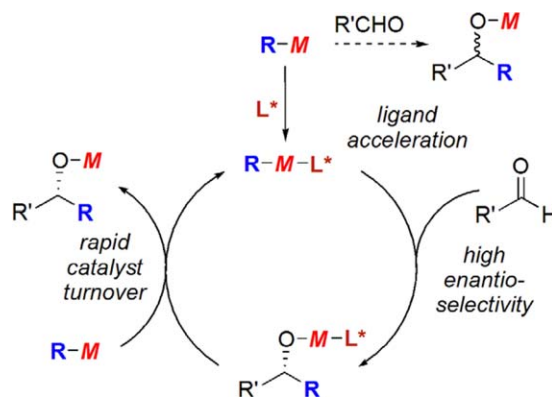
hols, they are activated to undergo enantioselective addition to aldehydes, thus realizing the catalytic enantioselective reaction.

Three general requirements were pointed out by Noyori to realize catalytic enantioselective addition of organometallic reagents (Scheme 2).^[6c] First, chiral ligand L* should have high selectivity to enable enantioselective addition of a chirally modified reagent R–M–L*. Second, the ligand should enhance significantly the reactivity of the modified reagent (ligand acceleration) in such a way that the racemic reaction by unmodified reagent R–M is negligible. Finally, at a catalyst turnover step, ligand L* should be transferred rapidly to R–M to regenerate R–M–L*. It has been demonstrated by mechanistic study that the DAIB-catalyzed reaction fulfills these three requirements well.^[6c]

Following the epoch-making discovery of the DAIB catalyst system, significant advances were made in the development



Scheme 1. DAIB-catalyzed enantioselective addition of Et₂Zn.



Scheme 2. General requirements for catalytic enantioselective addition of organometallic reagents.

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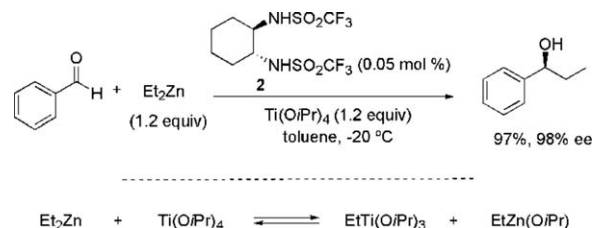
of the catalytic enantioselective carbonyl addition of organometallic reagents.^[7] According to the review article by Pu and Yu in 2001,^[7b] about 600 kinds of chiral catalysts were reported in the ten years prior. In most reports, diorganozinc reagents, especially Et_2Zn , are used.^[8] The zinc reagents are prepared by the reaction of the corresponding iodide (RI) with zinc followed by distillation (or sublimation) of the resulting RZnI to shift the Schlenk equilibrium. Accordingly, commercially available diorganozinc reagents are limited to volatile compounds.^[9] In retrospect, the attention of chemists shifted to the less reactive organometallic reagents from the conventional Grignard and organolithium reagents, which had earlier been studied extensively in the stoichiometric enantioselective addition.

1.2. Chiral Titanium Catalysts

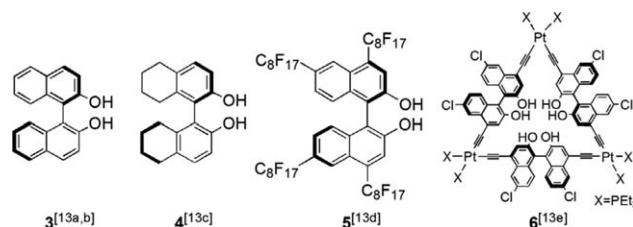
In 1989, enantioselective alkylation of a different type was reported by Yoshioka, Ohno, and co-workers (Scheme 3).^[10] The reaction was carried out with a mixed reagent of diethylzinc and $\text{Ti}(\text{O}i\text{Pr})_4$ in the presence of chiral bis-triflylamide **2**. The mixed reagent generates ethyltitanium species $\text{EtTi}(\text{O}i\text{Pr})_3$ under equilibrium, which undergoes enantioselective addition by a chiral titanium catalyst derived from **2** and $\text{Ti}(\text{O}i\text{Pr})_4$.^[11] High enantioselectivity was reported even at 0.05 mol % of **2**, demonstrating the high activity of the titanium catalyst.

The discovery of mixed reagents derived from dialkylzincs and $\text{Ti}(\text{O}i\text{Pr})_4$ led to the subsequent development of titanium catalysts derived from chiral diols such as TADDOLs^[12] and BINOLs.^[13] Taking advantage of the relative ease of modification of the steric and electronic nature, extensive work has been performed for the later ligands (Scheme 4).^[14]

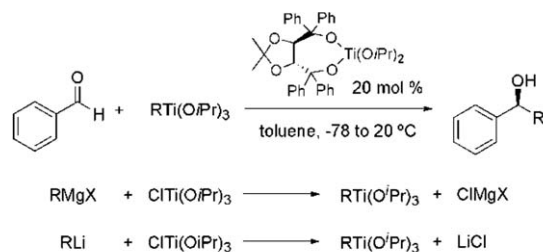
The use of organotitanium reagents themselves in the catalytic enantioselective addition was reported in 1994 by Seebach and co-worker.^[15] The titanium reagents were prepared by treatment of a Grignard reagent or lithium reagent with $\text{ClTi}(\text{O}i\text{Pr})_3$ and used after the strict removal of inorganic salts



Scheme 3. Chiral bis-triflylamide titanium catalyzed enantioselective addition of Et_2Zn .



Scheme 4. Chiral BINOL ligands for chiral titanium catalysts.



RM	removal of salts	yield (%)	ee(%)
EtMgBr	none	66	36
	dioxane/centrifugation	59	>98
BuLi	none	90	60
	12-crown-4/centrifugation	81	>98

Scheme 5. Enantioselective addition of organotitanium reagents catalyzed by TADDOL–titanium complex.

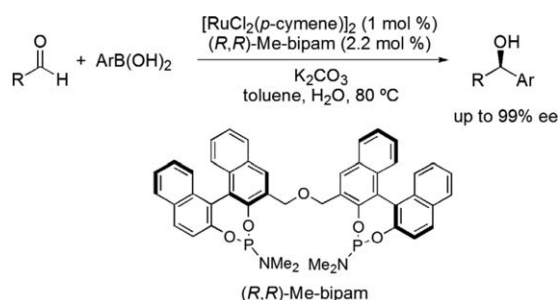
by precipitation and centrifugation (Scheme 5). Otherwise, enantioselectivities were degraded considerably.

1.3. Transition-Metal-Catalyzed Arylation with Arylboronic Acids

In 1998, Miyaura and co-workers reported the enantioselective arylation of aldehydes with arylboronic acids catalyzed by a chiral phosphine–Rh(I) complex.^[16] The major advantage of this reaction is the use of air- and moisture-stable boronic acids as arylating reagents. The reaction, involving other transition-metal catalysts, has been extensively improved in terms of catalyst efficiency, substrate scope, and reagents,^[17] and is now established as one of the most reliable methods for the enantioselective arylation of aldehydes (Scheme 6).^[18] It should be noted, however, that, in

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Scheme 6. Enantioselective arylation with arylboronic acids catalyzed by a chiral Ru complex.

spite of the commercial availability of a variety of arylboronic acids, they are produced in most instances by the borylation of aryl Grignard and lithium reagents.

1.4. Highly Active Catalyst System for Practical Application

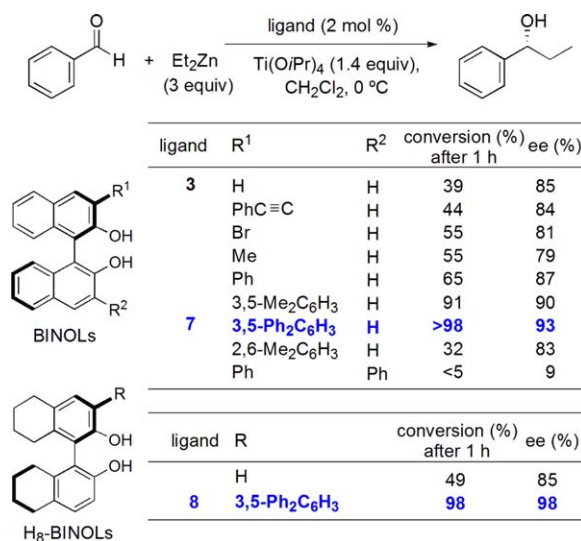
The catalytic enantioselective carbonyl addition reaction has long attracted the interest of chemists because of its synthetic importance. The studies in this area have resulted in the development of many highly enantioselective reactions. However, with few exceptions, the reactions are carried out at relatively high catalyst loadings (≥ 5 mol %), making them less practical for scale-up applications. In addition, organometallic reagents employed as the carbon nucleophiles have been limited to those with relatively low reactivity, such as diorganozincs and arylboronic acids. To enhance the practicality, it is necessary to develop reactions that can be carried out at lower catalyst loadings and utilize more conventional organometallic reagents of high reactivity, including Grignard and organolithium reagents. For such practical reactions, a chiral catalyst system with a superb turnover frequency is essential, that not only exhibits a high degree of ligand acceleration but also rapidly regenerates an active catalyst species at the turnover step, to overwhelm potential non-catalyzed background carbonyl addition at the low catalyst loading.

This account describes our recent development of a highly active chiral titanium catalyst system for the enantioselective carbonyl addition reaction to aldehydes by using a variety of organometallic reagents RM (M = MgX, Li, BY₂, ZnX, AlMe₂) at the lower catalyst loadings.

2. Catalytic Enantioselective Alkylation and Arylation of Aldehydes Using Grignard Reagents

2.1. Discovery of DPP-BINOL and DPP-H₈-BINOL

We found that the introduction of a substituent at the 3-position of BINOL and H₈-BINOL enhances significantly the activity of the resulting chiral titanium catalysts in the enantio-

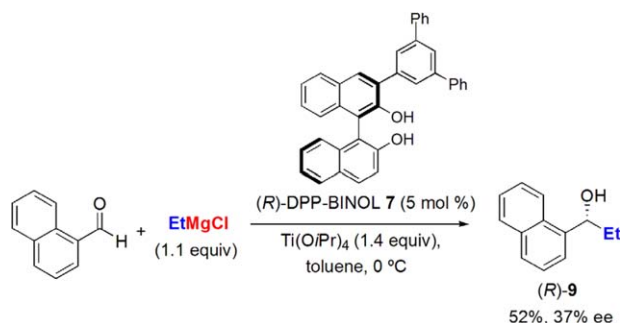


Scheme 7. Effects of substituents at the 3-position of BINOLs and H₈-BINOLs.

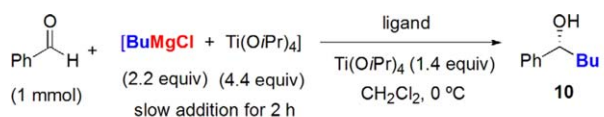
selective ethylation of aldehydes with the mixed reagent of Et₂Zn and Ti(OiPr)₄ (Scheme 7).^[19] Although the parent BINOL was reported to provide the ethylation product in 92% *ee* at 20 mol % loading,^[13b] the reaction employing 2 mol % of the ligand did not reach full conversion even after 5 h and the product was obtained in 85% *ee*. The decrease in the enantioselectivity implies that background direct addition of EtTi(OiPr)₃ proceeded concurrently with the ligand-accelerated enantioselective reaction. Under similar conditions, the reaction with DPP-BINOL **7** (DPP = 3,5-diphenylphenyl) was accomplished within 1 h to give the product in 93% *ee*, suggesting the high activity of the titanium catalyst derived from **7**. Examination of a series of BINOL derivatives revealed that (i) the catalytic activity, assessed by the conversion after 1 h, is increased by the increase of the size of the substituent at the 3-position of the ligands (H < PhC≡C < Br ≈ Me < Ph, 3,5-Me₂-C₆H₃ < 3,5-Ph₂-C₆H₃); (ii) the enantioselectivity is increased almost in this order; (iii) the introduction of the 2,6-Me₂-C₆H₃ group of excessive steric hindrance results in a significant decrease in the catalytic activity; (iv) catalytic activity is almost lost by the introduction of the two phenyl groups at the 3- and 3'-positions symmetrically; and (v) enhancement in catalytic activity is also observed for H₈-BINOL derivative **8** (DPP-H₈-BINOL), which exhibits the highest selectivity of 98% *ee* at 2 mol % in these ligands.

2.2. Challenge: Reaction with Alkyl Grignard Reagents

The discovery of highly active chiral titanium catalyst systems derived from DPP-BINOL **7** and DPP-H₈-BINOL **8** prompted us to investigate enantioselective additions using Grignard reagents. We focused our attention on the use of a



Scheme 8. Initial attempt at catalytic enantioselective alkylation using Grignard reagents.



ligand (amount)	yield (%)	ee (%)
DPP-BINOL 7 (2 mol %)	86	93
DPP-BINOL 7 (2 mol %)	94	92 ^[a]
DPP-BINOL 7 (4 mol %)	78	97 ^[b]
DPP-H ₈ -BINOL 8 (2 mol %)	86	92
DPP-H ₈ -BINOL 8 (4 mol %)	83	97 ^[b]
BINOL 3 (2 mol %)	78	79

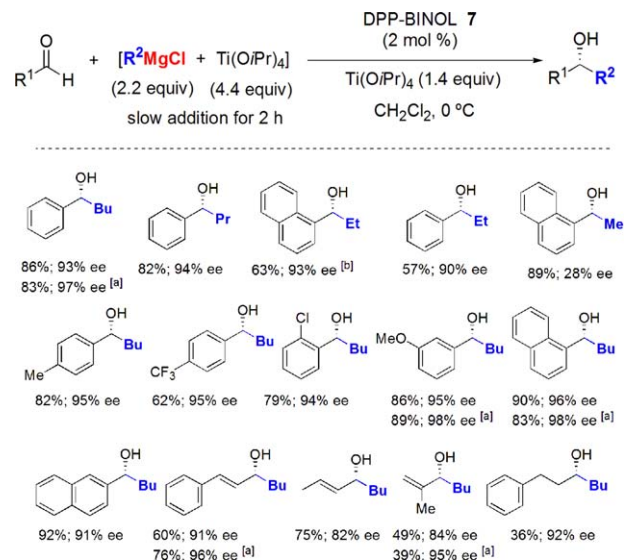
[a] PhCHO (10 mmol) was used. [b] Slow addition for 3 h.

Scheme 9. Enantioselective butylation of benzaldehyde catalyzed by titanium complexes derived from DPP-BINOL 7 and DPP-H₈-BINOL 8.

mixed reagent of Grignard reagents and Ti(OiPr)_4 , which might produce organotitanates $\text{R-Ti(OiPr)}_4\text{-MgX}^{[20]}$ without formation of the undesirable Lewis acidic inorganic salts.

As an initial attempt, we examined the reaction of 1-naphthaldehyde and EtMgCl (Scheme 8). When the reaction was carried out by adding the Grignard reagent to a toluene solution of the aldehyde, (R)-DPP-BINOL 7 (5 mol %), and Ti(OiPr)_4 (1.4 equiv), the ethylation product (R)-9 was obtained in 52% yield and 37% ee. The absolute structure of the major enantiomer was the same as that observed in the reaction with Et_2Zn . In view of the likely background direct Grignard reaction, the observed enantioselectivity, although relatively low, encouraged us to optimize the reaction conditions.

The following optimization study led us to establish reaction conditions that enable high enantioselectivity at 2 mol % catalyst loading, as exemplified by the butylation of benzaldehyde using BuMgCl (2 M in Et_2O) (Scheme 9).^[21] Thus, the Grignard reagent is first mixed with Ti(OiPr)_4 in CH_2Cl_2 and the resulting mixed reagent is added for 2 h to a CH_2Cl_2 solution of the aldehyde, Ti(OiPr)_4 , and 2 mol % of DPP-BINOL 7 at 0 °C. Additional stirring for 1 h gave the butylation product 10 in 86% yield and 93% ee. The reaction can be carried



[a] DPP-H₈-BINOL 8 (4 mol %) was used. [b] EtMgBr was used.

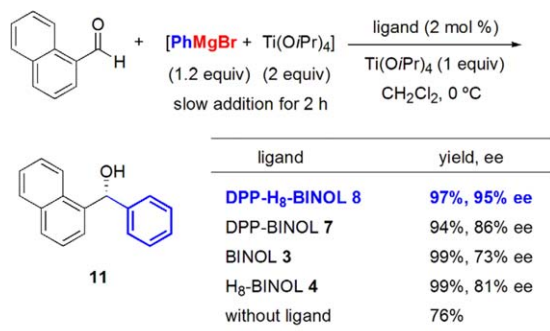
Scheme 10. Scope and limitations of the catalytic enantioselective alkylation using Grignard reagents.

out on a 10 mmol scale without difficulty. DPP-H₈-BINOL 8 also exhibits high selectivity comparable to DPP-BINOL 7. On the other hand, the parent BINOL 3 does not show high enantioselectivity.^[22] The enantioselectivity could be further improved to 97% ee by using 4 mol % of the DPP ligands 7 and 8 with 3 h addition times.^[23]

The scope and limitations of the enantioselective alkylation using Grignard reagents are summarized in Scheme 10. The reactions of primary alkyl Grignard reagents proceed with high enantioselectivity to give the corresponding alkylation products. The yields of the ethylation products are moderate, owing to the lower stability of the mixed reagent derived from EtMgX ($\text{X} = \text{Cl}, \text{Br}$). A mixed titanium reagent derived from MeMgCl undergoes smooth addition, but exhibits poor enantioselectivity. Mixed titanium reagents derived from *sec*- and *tert*-alkyl Grignard reagents decompose rapidly and cannot be used. The reaction is applicable to a range of benzaldehyde derivatives, either with an electron-donating or electron-withdrawing substituent at the para, meta, and ortho positions, to afford the corresponding alkylation products in high yields and high enantioselectivities. The reactions of α,β -unsaturated aldehydes also proceed in an efficient manner to give the corresponding allylic alcohols with high enantioselectivity. Aliphatic aldehydes are less reactive, affording the corresponding products in lower yields but with high selectivities.

2.3. Reaction with Aryl Grignard Reagents

The approach based on the use of Grignard reagents as the mixed reagents with Ti(OiPr)_4 is also successful in the



Scheme 11. Enantioselective phenylation of 1-naphthaldehyde catalyzed by the titanium complex derived from DPP-H₈-BINOL **8**.

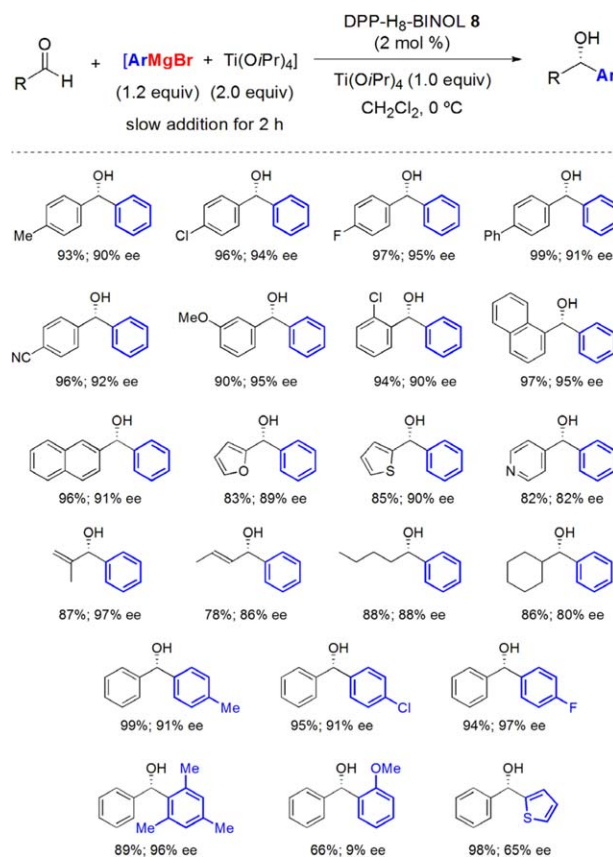
enantioselective arylation of aldehydes.^[21b,24] The mixed reagents derived from aryl Grignard reagents are stable and do not need to be employed in excess. Thus, under optimized conditions with DPP-H₈-BINOL **8** (2 mol %), the reaction of 1-naphthaldehyde with the mixed reagent of PhMgBr (3 M in Et₂O, 1.2 equiv) and Ti(OiPr)₄ (2 equiv) affords addition product **11** in 97% yield and 95% ee (Scheme 11). For the arylation reaction, DPP-H₈-BINOL **8** exhibits superior selectivity in comparison with DPP-BINOL **7** and the parent ligands **3** and **4**. In the absence of the catalyst, the mixed titanium reagent undergoes rapid racemic phenylation, implying that the catalytic enantioselective phenylation is very fast, overwhelming the background reaction.

The enantioselective arylation reaction shows wide generality both for aldehydes and for Grignard reagents, realizing easy access to the enantiomerically enriched secondary alcohols (Scheme 12). Aromatic as well as heteroaromatic, unsaturated, and aliphatic aldehydes are good substrates for this reaction. Various commercially available aryl Grignard reagents can be employed to give the corresponding arylation products in high ee, except for the *o*-methoxyphenyl derivative. Enantioselectivity is moderate for the reaction with the 2-thienyl Grignard reagent.

Application of the reaction conditions optimized for aryl Grignard reagents to aryllithium reagents results in the poor enantioselectivity of the arylation products. However, high enantioselectivity can be achieved when the arylation of aldehydes is carried out with a mixed reagent of an aryllithium (1.5 equiv), Ti(OiPr)₄ (2.5 equiv), and MgBr₂ (1.5 equiv) in the presence of DPP-H₈-BINOL (2 mol %).^[23c,25]

2.4. Reaction with Functionalized Aryl Grignard Reagents Generated In Situ from Aryl Halides

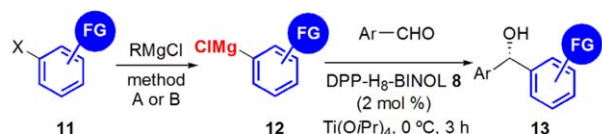
The scope of the catalytic enantioselective arylation reaction was expanded further by the use of functionalized Grignard reagents **12**, which are prepared in situ by halogen/Mg



Scheme 12. Scope and limitations of the catalytic enantioselective arylation using Grignard reagents.

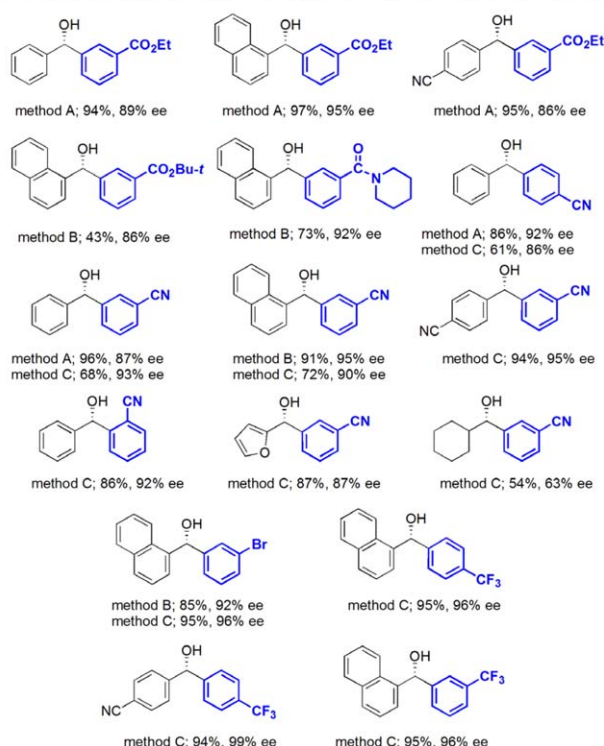
exchange of aryl halides **11** (Scheme 13).^[21b,26] In method A, the functionalized Grignard reagents are generated by I/Mg exchange reaction of aryl iodides with *i*PrMgCl in THF according to a protocol developed by Knochel et al.^[27] The resulting reagents are used in the subsequent enantioselective reaction after removing THF in vacuo, because otherwise the enantioselectivity of the reaction is decreased considerably. Alternatively, the Grignard reagents can be generated in Et₂O by using *c*-C₅H₉MgCl in the I/Mg exchange reaction. The resulting reagents can be employed in the enantioselective addition to aldehydes without exchanging the solvent system (method B). Grignard reagents bearing functional groups, such as ester/amide carbonyls, cyano groups, or bromine atoms, afford the corresponding functionalized diarylmethanols **13** with high chemical and optical yields.

Aryl bromides would be preferable precursors for functionalized Grignard reagents in light of their stability, availability, and low cost. Although Br/Mg exchange of aryl bromides is a less efficient process with *i*PrMgCl, the turbo Grignard reagents *i*PrMgCl·LiCl of enhanced reactivity have been reported to undergo the exchange reaction.^[28] The resulting



For iodides (X = I):
method A: 1) *i*PrMgCl, THF, -40 °C, 2) Ti(O*i*Pr)₄, 3) *in vacuo*, 4) Et₂O, CH₂Cl₂
method B: 1) *n*-C₅H₉MgCl, Et₂O, -20 °C, 2) Ti(O*i*Pr)₄, CH₂Cl₂

For bromides (X = Br):
method C: 1) *i*PrMgCl·LiCl, THF, rt, 2) Ti(O*i*Pr)₄, 3) *in vacuo*, 4) Et₂O, CH₂Cl₂

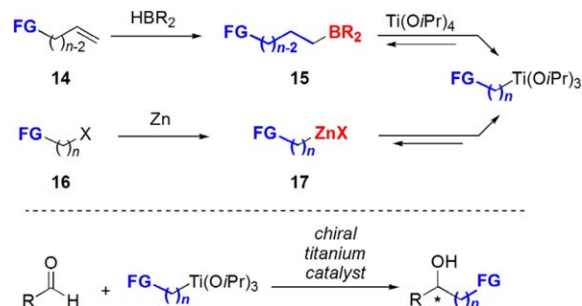


Scheme 13. Catalytic enantioselective addition of functionalized aryl groups using Grignard reagents generated *in situ* from aryl halides.

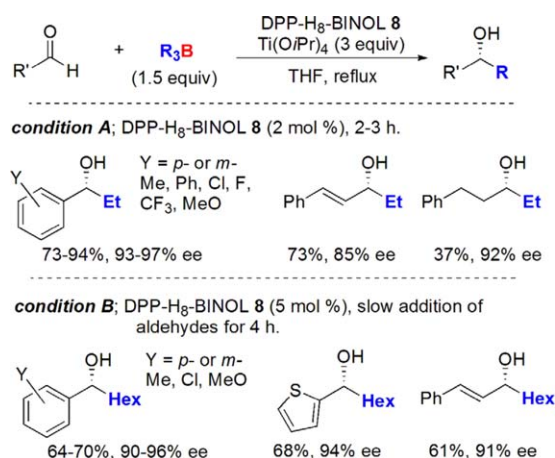
aryl Grignard reagents, complexed with LiCl, can also be employed successfully in the enantioselective addition to aldehydes (method C).^[26] Beginning with commercially available aryl bromides and aldehydes, a variety of chiral secondary alcohols were prepared with high enantioselectivity. This method is applicable to aryl bromides bearing a CN group, a CF₃ group, or a bromine atom, while carbonyl functionalities do not tolerate these conditions.

3. Catalytic Enantioselective Addition of Functionalized Alkylmetal Reagents

Recent advances in functionalized organometallic reagents have considerably increased the scope of carbon nucleophiles,



Scheme 14. Catalytic enantioselective addition of functionalized alkylmetal reagents.

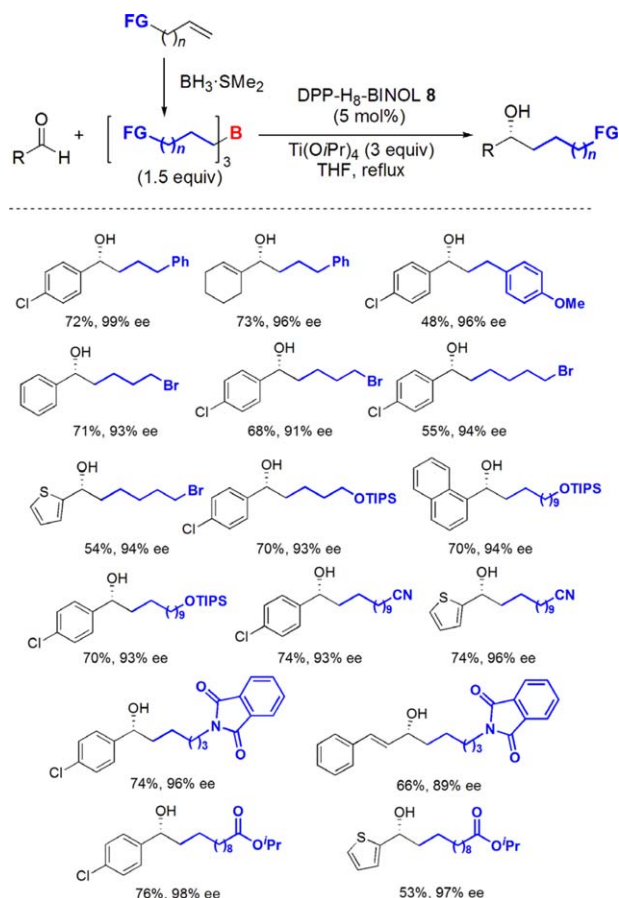


Scheme 15. Catalytic enantioselective alkylation with a mixed reagent of trialkylborane and Ti(O*i*Pr)₄.

allowing short and efficient syntheses of polyfunctional target molecules without unproductive protection/deprotection steps.^[29] Despite its potential utility in asymmetric synthesis, the catalytic enantioselective addition of functionalized alkylmetal reagents to aldehydes is rare.^[9,30] To realize such transformations, we focused our attention on alkylboron reagents **15** and alkylzinc halide reagents **17** because of their excellent functional group tolerance and easy preparation from terminal alkenes **14** and haloalkanes **16**, respectively (Scheme 14). Although there had not been any precedent, we anticipated that transmetalation of these functionalized reagents with Ti(O*i*Pr)₄ would generate the corresponding functionalized alkyltitanium species *in situ*, undergoing enantioselective addition to aldehydes in the presence of 3-substituted BINOL derivatives.

3.1. Reaction Utilizing Functionalized Alkylboron Reagents

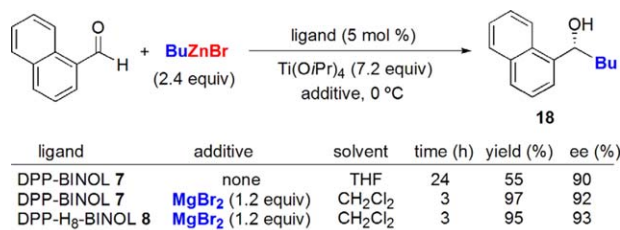
We found that Et₃B undergoes enantioselective addition to aldehydes in the presence of Ti(O*i*Pr)₄ (3 equiv) and DPP-H₈-



Scheme 16. Catalytic enantioselective addition of functionalized alkyl groups using boron reagents generated from terminal alkenes.

BINOL **8** (2 mol %) in THF at reflux temperature (Scheme 15, condition A).^[31] Although the reaction is carried out at a relatively high temperature, enantioselectivity is high for aromatic, α,β -unsaturated, and aliphatic aldehydes. Unfortunately, application of condition A to the higher alkylboron reagents such as Hex₃B resulted in low yields of the alkylation products owing to the competing hydride reduction by the boron reagents through a dehydroboration mechanism. However, the yields of alkylation products could be improved without the loss of high enantioselectivity by carrying out the reaction with slow addition of aldehydes (condition B).^[32]

The results of control experiments suggest that trialkylborane R₃B undergoes transmetalation with Ti(OiPr)₄ to form alkyltitanium species RTi(OiPr)₃, or its aggregate with co-generated R₂B(OiPr), as the active alkylating reagent.^[33] Although the transmetalation proceeds at temperatures higher than those for the zinc and aluminum reagents,^[34] the highly active DPP-H₈-BINOL-derived catalyst provides a unique opportunity to utilize alkylboron reagents in combination with Ti(OiPr)₄.



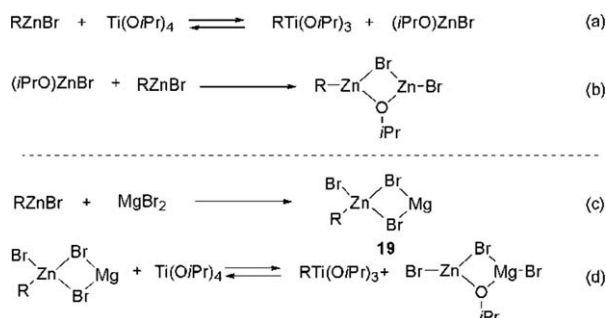
Scheme 17. Catalytic enantioselective butylation of 1-naphthaldehyde using BuZnBr.

Under condition B, functionalized alkylboron reagents, prepared by hydroboration of terminal alkenes with BH₃-SMe₂, undergo enantioselective addition to aldehydes to give the corresponding functionalized secondary alcohols with high enantioselectivity (Scheme 16). The reaction tolerates various functionalities including bromine, TIPS-protected hydroxy, phthalimide-protected amino, *i*-propyl ester, and cyano groups, providing a useful entry to chiral building blocks.

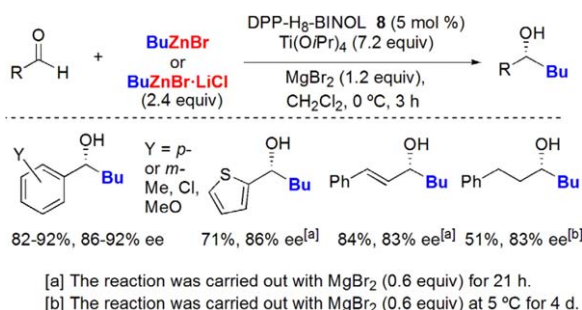
3.2. Reaction Utilizing Functionalized Alkylzinc Bromides

An organozinc reagent is another candidate for the catalytic enantioselective addition of functionalized alkyl groups to aldehydes. Of the several types of zinc reagents, we focused on functionalized alkylzinc bromides FG-(CH₂)_n-ZnBr because of their superb functional group tolerance and easy preparation.^[29] Importantly, the zinc reagents are prepared in an atom-economical manner by the reaction of bromide precursors with active Rieke zinc^[35] or, more conveniently, zinc dust in the presence of LiCl.^[36] Taking advantage of their intrinsic low reactivity, organozinc halides have been utilized in the formation of carbon-carbon bonds by transition-metal catalysis, as represented by the Negishi cross-coupling reaction.^[37] However, their application in catalytic enantioselective carbonyl additions has been hampered by their low reactivity.^[9i,38]

To realize the reaction using alkylzinc bromides, we first examined the butylation of 1-naphthaldehyde using a commercially available THF solution of BuZnBr under conditions similar to those for Grignard reagents (Scheme 17). In the presence of DPP-BINOL **7** (5 mol %), the mixed reagent of BuZnBr and Ti(OiPr)₄ underwent addition to the aldehyde to give the butylation product **18** in 90% ee. However, the reaction slowed down at lower conversion and the chemical yield was moderate even after 24 h. To improve conversion, we surveyed various metal salts and found MgBr₂ to be an effective additive. Thus, in the presence of MgBr₂ (1.2 equiv) in CH₂Cl₂, the butylation reaction is complete within 3 h without the loss of enantioselectivity. Under these conditions,



Scheme 18. Hypothetical transmetalation via Zn–Mg dinuclear complex **19**.



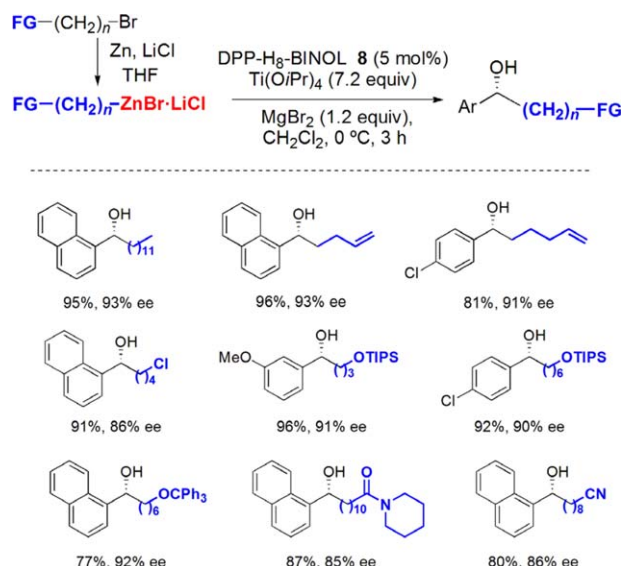
Scheme 19. Chiral titanium-catalyzed enantioselective butylation using BuZnBr.

DPP-H₈-BINOL **8** also exhibits high chemical yield and enantioselectivity.^[39]

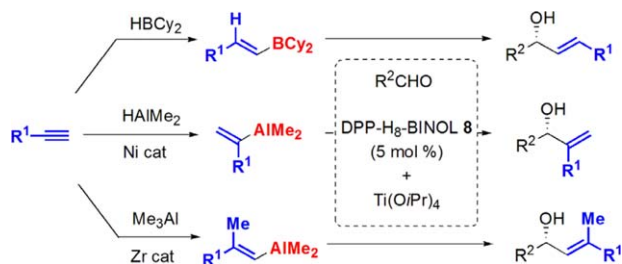
Assuming that transmetalation of an alkylzinc bromide with $\text{Ti}(\text{O}i\text{Pr})_4$ generates an alkyltitanium reagent for the catalytic enantioselective addition (Scheme 18, equation (a)), a possible interpretation for the lower conversion in the absence of the additive is that the zinc reagent is consumed simultaneously by aggregation with co-produced $(i\text{PrO})\text{ZnBr}$ (equation (b)). According to a recent structural study on Zn–Mg dinuclear complexes,^[40] MgBr_2 as an additive is likely to form a dinuclear complex (**19**) of higher reactivity (equation (c)), which reacts with $\text{Ti}(\text{O}i\text{Pr})_4$ to generate the alkyltitanium reagent in a more efficient manner (equation (d)).

Under the optimized conditions with DPP-H₈-BINOL **8** (5 mol %), a variety of aromatic aldehydes undergo highly enantioselective butylation to give the products in high yields (Scheme 19). For heteroaromatic and unsaturated aldehydes, satisfactory results are obtained with 0.6 equivalents of MgBr₂. The reactions of aliphatic aldehydes are slow but enantioselective.^[39]

The scope of the reaction with respect to the functionalized zinc reagents is summarized in Scheme 20. The zinc reagents are prepared from bromide precursors and zinc dust in the presence of LiCl and then subjected to the reaction of aromatic aldehydes to obtain the corresponding functionalized alcohols in high yields and high enantioselectivities. Functional groups, such as



Scheme 20. Catalytic enantioselective addition of functionalized alkyl groups using zinc reagents generated from bromide precursors.

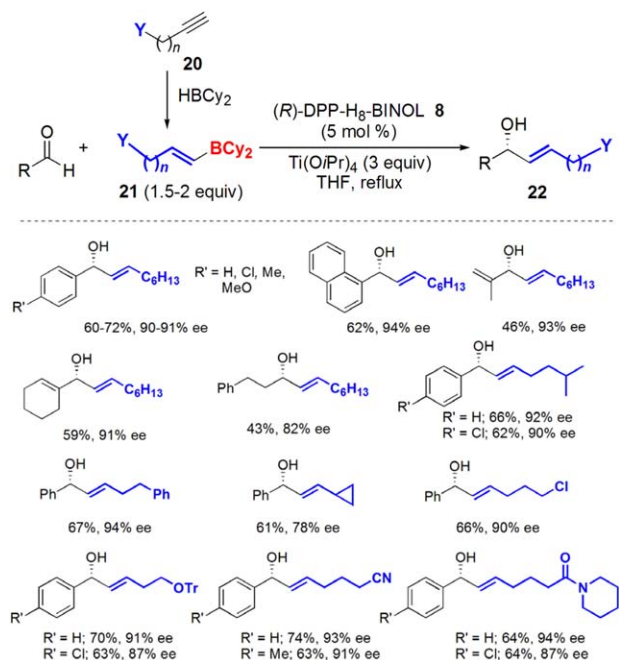


Scheme 21. Catalytic enantioselective synthesis of secondary allylic alcohols beginning with terminal alkynes and aldehydes.

chlorine, silyl- and trityl-protected hydroxy, amide, and cyano groups, are tolerated in this reaction.

4. Catalytic Enantioselective Addition of Vinylmetal Reagents Generated from Terminal Alkynes

Enantiomerically enriched secondary allylic alcohols are versatile chiral building blocks in organic synthesis. In addition, a variety of highly diastereoselective reactions can be used to convert them into further elaborated building blocks. There is currently an increasing interest in catalytic enantioselective methods for the synthesis of chiral allylic alcohols with various substitution patterns.^[7i] The chiral titanium catalyst system provided an opportunity to develop an efficient method for the enantioselective synthesis of secondary allylic alcohols with different substitution patterns starting from terminal alkynes and aldehydes (Scheme 21).



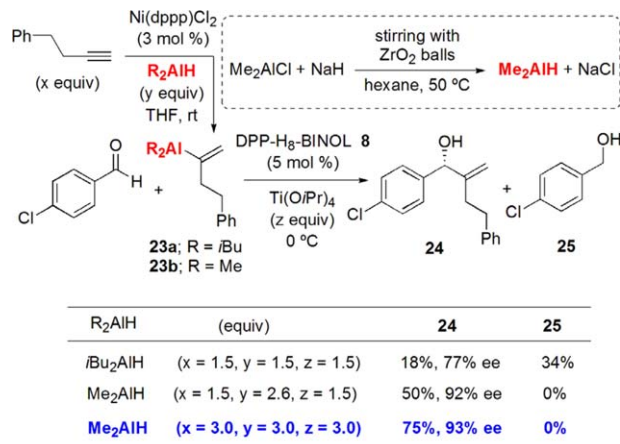
Scheme 22. Catalytic enantioselective synthesis of β -substituted allylic alcohols beginning with terminal alkynes.

4.1. *trans*- β -Substituted Allylic Alcohols by Enantioselective Addition of Vinylboron Reagents

Hydroboration of 1-octyne with Cy_2BH in THF and treatment of the resulting vinylboron reagent **21** ($\text{Y} = \text{H}$, $n = 6$) with aromatic aldehydes in the presence of DPP-H₈-BINOL **8** (5 mol %) and $\text{Ti}(\text{O}i\text{Pr})_4$ (3 equiv) at 66 °C for 2 h affords vinylation products **22** ($\text{Y} = \text{H}$, $n = 6$) in 60–72% yields with high enantioselectivities (90–94% *ee*) (Scheme 22).^[41] No cyclohexylation product is observed, implying the selective transfer of the vinyl group to the titanium atom. The vinylboron reagent also reacts with α,β -unsaturated aldehydes to give the corresponding divinyl carbinols with high enantioselectivity. Lower yet still acceptable enantioselectivity is obtained for aliphatic aldehydes. The reaction can start with a variety of terminal alkynes **20**, including those containing a chlorine atom, a protected alcohol, a nitrile, and an amide group, leading to the enantioselective formation of the corresponding functionalized *trans*- β -substituted allylic alcohols **22**.

4.2. α -Substituted Allylic Alcohols by Enantioselective Addition of Vinylaluminum Reagents

Recently, Hoveyda and co-workers have developed a method for the generation of α -substituted vinylaluminum reagents by a Ni-catalyzed Markovnikov-type hydroalumination of terminal alkynes with $i\text{Bu}_2\text{AlH}$.^[42] To develop a catalytic enantioselective synthesis of chiral α -substituted allylic alcohols, we

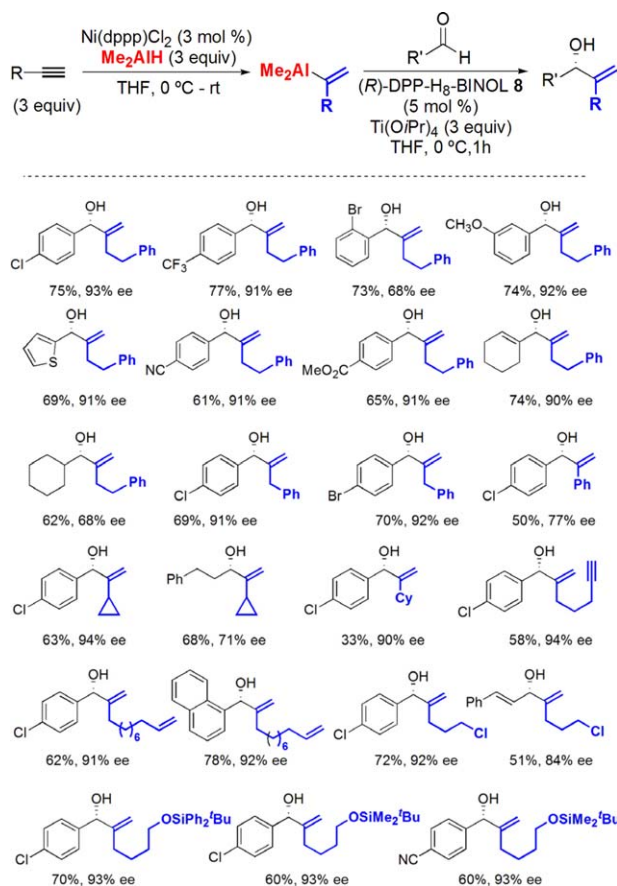


Scheme 23. Catalytic enantioselective vinylation using aluminum reagents generated from 4-phenylbut-1-yne.

applied the chiral titanium catalyst system to the addition of thus-generated α -substituted vinylaluminum reagents.^[43,44]

In our initial attempt, α -substituted vinylaluminum reagent **23a** was prepared by the reaction of 4-phenylbut-1-yne with $i\text{Bu}_2\text{AlH}$ according to the reported protocol^[42] and treated with *p*-chlorobenzaldehyde in the presence of $\text{Ti}(\text{O}i\text{Pr})_4$ and DPP-H₈-BINOL **8** (5 mol %) (Scheme 23). The anticipated vinylation product **24** was obtained in good *ee*. However, the chemical yield was poor and reduced alcohol **25** was a major product. The vinylaluminum reagent **23a** bearing isobutyl groups is likely to undergo a preferential hydride reduction of the aldehyde. This problem could be overcome by the use of Me_2AlH in the hydroalumination. Thus, vinylaluminum reagent **23b**, generated by using Me_2AlH , underwent enantioselective addition to the aldehyde without formation of byproduct **25**. Under the optimized conditions employing three equivalents each of the alkyne, Me_2AlH , and $\text{Ti}(\text{O}i\text{Pr})_4$, **24** was obtained in 75% yield and 93% *ee*. It is noteworthy that Me_2AlH (ca. 1.1 M solution in hexane) can be prepared readily from Me_2AlCl and NaH. Although the reaction is quite sluggish under conventional stirring or under sonication, quantitative formation of Me_2AlH is achieved by stirring the reaction mixture in the presence of zirconia (ZrO_2) balls. The remarkable effect of zirconia balls might be the consequence of the mechanical removal of NaCl covering the surface of NaH.

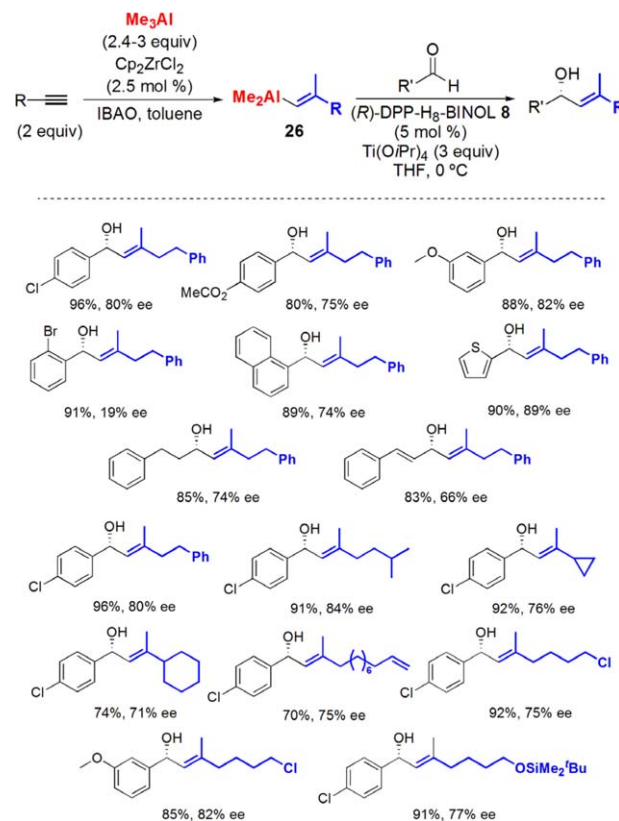
As illustrated in Scheme 24, a variety of α -substituted allylic alcohols can be synthesized in high enantiopurity starting from terminal alkynes and aldehydes through the hydroalumination/catalytic addition sequence. The reaction is applicable to a variety of terminal alkynes including an enyne and a diyne. Notably, the reaction tolerated potentially reactive cyano and ester groups in aldehydes and chlorine atoms and silyloxy groups in alkynes.



Scheme 24. Catalytic enantioselective synthesis of α -substituted allylic alcohols beginning with terminal alkynes.

4.3. β,β -Disubstituted Allylic Alcohols by Enantioselective Addition of Vinylaluminum Reagents

The Zr-catalyzed carboalumination of terminal alkynes has been well established as a reliable method for the generation of β,β -disubstituted vinylaluminum reagents of *E* geometry.^[45] Despite their accessibility, the vinylaluminum reagents had not been used in the enantioselective addition to aldehydes, probably because of their high reactivity that degrades the selectivity through non-catalyzed background reaction. Taking advantage of the high activity of the DPP-H₈-BINOL-derived titanium catalyst system, a simple one-pot method was developed beginning with terminal alkynes, including those bearing an olefinic moiety, a chlorine atom, and a silyloxy group, and aromatic or heteroaromatic aldehydes (Scheme 25).^[46] Thus, disubstituted vinylaluminum reagent **26** is prepared by the Zr-catalyzed carboalumination of a terminal alkyne with Me₃Al and treated with an aldehyde in the presence of Ti(OiPr)₄ and DPP-H₈-BINOL **8** (5 mol %) at 0 °C in THF. The vinylation reaction is generally accomplished within 30 min to provide the corresponding β,β -disubstituted allylic alcohol in high yield and in



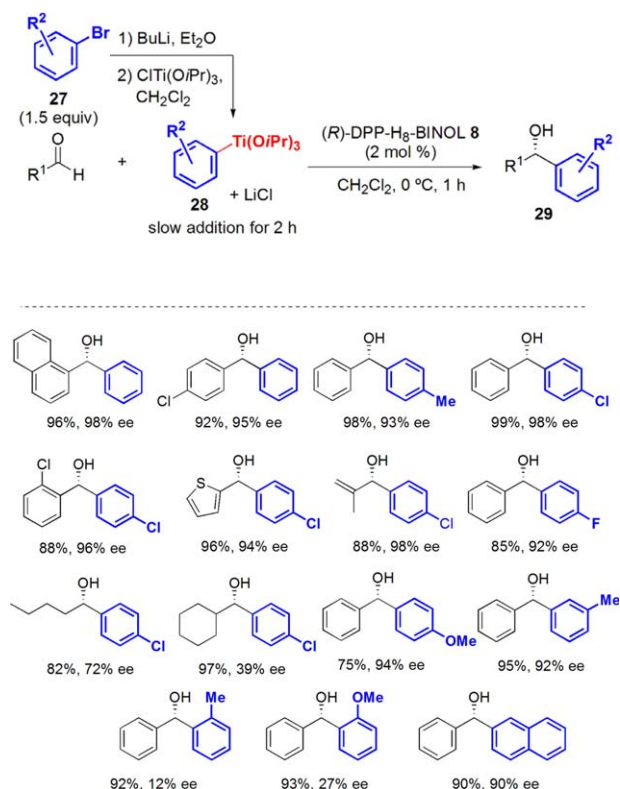
Scheme 25. Catalytic enantioselective synthesis of β,β -disubstituted allylic alcohols beginning with terminal alkynes.

good to high enantioselectivity. The reaction is carried out at relatively dilute conditions (0.01 M for aldehydes) to suppress the background reaction.

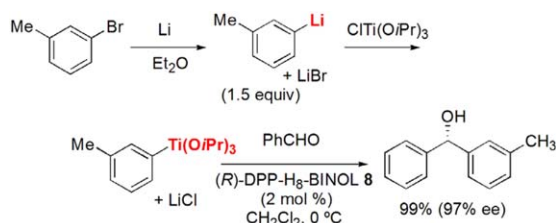
5. Catalytic Enantioselective Addition of Aryl- and Heteroaryl-titanium Reagents Generated In Situ via Lithium Reagents

5.1. Reaction with Aryllithium Reagents

The organotitanium reagent RTi(OiPr)₃ is the most straightforward reagent for the titanium-catalyzed enantioselective addition to aldehydes. Indeed, enantioselective addition of organotitanium reagents to aldehydes catalyzed by chiral titanium complexes derived from TADDOLs (10–20 mol %)^[15] and H₈-BINOL (5–10 mol %)^[47] has been reported previously. However, as described before, the titanium reagents, prepared from conventional organometal reagents RM (M = MgX, Li) and ClTi(OiPr)₃, are inevitably contaminated with Lewis acidic inorganic salts, which are detrimental to the enantioselectivity and therefore need to be removed rigorously. Hence, the tedious isolation process makes the reaction of the titanium reagents less practically attractive. We found that



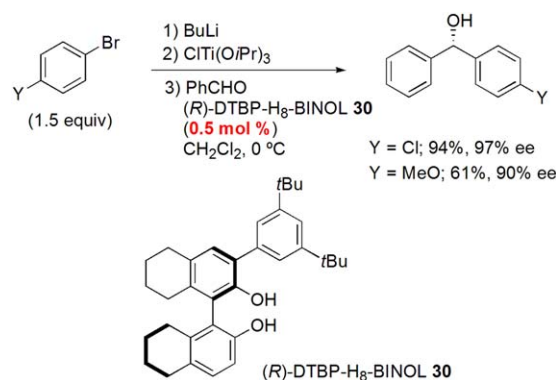
Scheme 26. Catalytic enantioselective addition of in situ generated aryltitanium reagents **28**.



Scheme 27. Enantioselective synthesis of phenyl-*m*-tolylmethanol.

aryltitanium reagents **28**, generated in situ from aryllithiums and $\text{ClTi}(\text{O}i\text{Pr})_3$, can be employed without removal of LiCl in the enantioselective arylation of aldehydes by virtue of the superior activity of the 3-substituted H_8 -BINOL-titanium catalysts (Scheme 26). Taking further advantage of the efficient generation of aryllithiums from aryl bromides by Br/Li exchange with BuLi , the reaction provides us with a practical means for the synthesis of enantioenriched secondary alcohols starting from commercially available aryl bromides and aldehydes.^[48,49]

The reaction begins with aryl bromides **27**, from which aryltitanium reagents **28** are prepared by lithiation with BuLi and transmetalation of the resulting aryllithium reagents with $\text{ClTi}(\text{O}i\text{Pr})_3$. In the presence of 2 mol % of (*R*)-DPP- H_8 -BINOL **8**, the titanium reagents undergo enantioselective



Scheme 28. Catalytic enantioselective arylation with 0.5 mol % of DTBP- H_8 -BINOL **30**.

addition to aldehydes to furnish the corresponding secondary alcohols **29**. The reactions of para- or meta-substituted phenyl bromides and 2-naphthyl bromide afford the corresponding benzaldehyde adducts in high yields and high selectivities (90–98% *ee*). The reaction is readily applicable to a variety of aromatic, heteroaromatic, and α,β -unsaturated aldehydes. Enantioselectivity is low to moderate for aliphatic aldehydes.

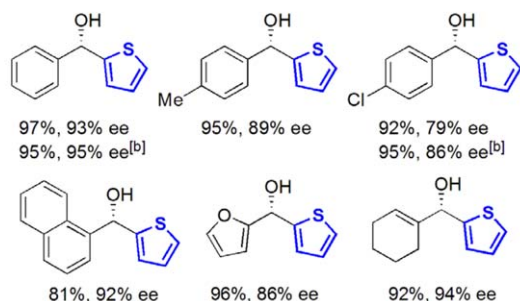
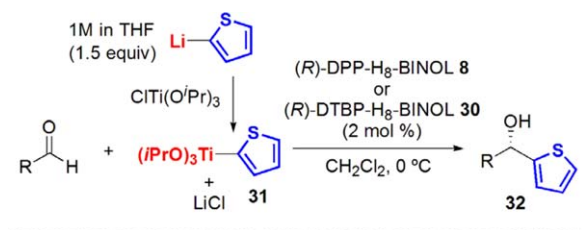
The arylation reaction can be readily carried out with a lithium reagent prepared from aryl bromides and metallic lithium without rigorous removal of co-produced LiBr , as exemplified by the enantioselective synthesis of phenyl-*m*-tolylmethanol (Scheme 27).

Screening of other H_8 -BINOL derivatives revealed that DTBP- H_8 -BINOL **30**, bearing the sterically more demanding 3,5-di-*tert*-butylphenyl group at the 3-position, also exhibits remarkable catalytic activity, even higher than that of DPP derivative **8**. Of practical importance, the arylation reaction can be carried out with 0.5 mol % of **30**, providing diarylmethanols with high enantioselectivity (Scheme 28).

5.2. Reaction with Heteroaryllithium Reagents

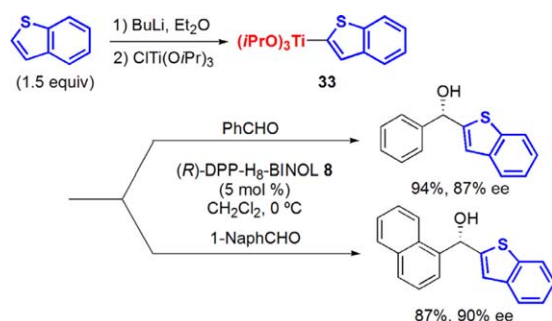
The importance of heteroaromatic frameworks in medicinal chemistry warrants the development of a catalytic method for the enantioselective addition of heteroaryl groups to carbonyl compounds.^[50] Preparation of a variety of heteroaryllithiums is well established by lithiation of the parent heterocycles or the halide derivatives. The thus-prepared heteroaryl reagents can be employed in enantioselective addition in the presence of the 3-aryl H_8 -BINOL-titanium catalysts.^[48]

In the presence of DPP- or DTBP- H_8 -BINOL (2 mol %), the reaction of benzaldehyde with 2-thienyllithium reagent **31**, generated in situ from commercially available 2-thienyllithium (1 M in THF), affords the phenyl(2-thienyl)-methanol **32** ($\text{R} = \text{Ph}$) in high yield and high *ee* (Scheme 29). Under these conditions, aromatic, heteroaryl, and α,β -unsaturated aldehydes undergo enantioselective addition of the



[a] Unless otherwise noted, DPP-H₈-BINOL **8** was used. [b] DTBP-H₈-BINOL **30** was used.

Scheme 29. Catalytic enantioselective addition of the 2-thienyl group to aldehydes.^[a]

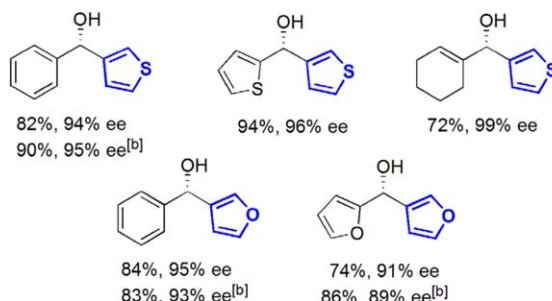
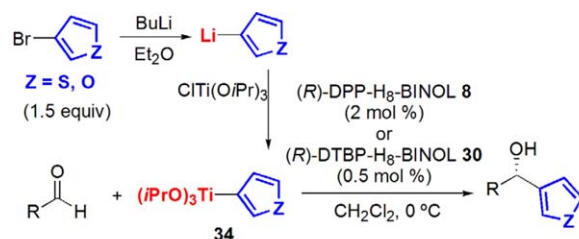


Scheme 30. Catalytic enantioselective addition of the 2-benzothiophenyl group to aldehydes.

heteroaryl titanium reagent. The enantioselective addition of the 2-benzothiophenyl group is also achievable by employing the corresponding heteroaryl titanium reagent **33** generated in situ from benzothiophene (Scheme 30).

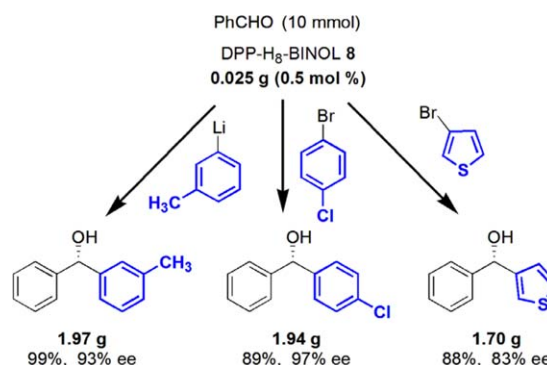
Enantioselective addition of 3-thienyl and 3-furyl groups to aromatic and unsaturated aldehydes can be achieved as well, beginning with the corresponding heteroaromatic bromide precursors via heteroaryl titanium reagents **34** (Scheme 31). DPP-H₈-BINOL **8** (2 mol %) exhibits high enantioselectivity in this reaction. Reaction at 0.5 mol % catalyst loading is possible with DTBP-H₈-BINOL **30**.

Of practical importance, the enantioselective arylation and heteroarylation reactions are operationally simple and can be carried out on a 10 mmol scale without any difficulty at 0.5 mol % catalyst loading (Scheme 32). Even with 25 mg of



[a] Unless otherwise noted, (R)-DPP-H₈-BINOL **8** (2 mol %) was used. [b] (R)-DTBP-H₈-BINOL **30** (0.5 mol %) was used.

Scheme 31. Catalytic enantioselective addition of 3-thienyl and 3-furyl groups to aldehydes.^[a]



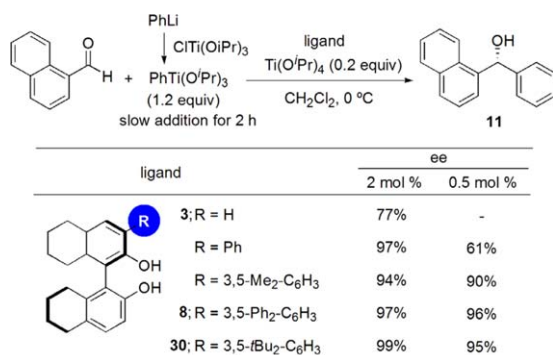
Scheme 32. Application to a gram-scale synthesis of enantioenriched diaryl- and aryl heteroaryl methanols.

(R)-DPP-H₈-BINOL, the enantioenriched diaryl- and aryl heteroaryl methanols can be synthesized in a gram quantity.

6. High Catalytic Activity

6.1. Evaluation of 3-Aryl H₈-BINOLs

As described in the preceding sections, introduction of the sterically demanding aryl group at the 3-position remarkably enhances the activity of BINOL- and H₈-BINOL-derived titanium catalysts. Since its discovery, we have been interested in elucidating the chemistry underlying the phenomenon. The excellent catalytic activity not only permits the reduction of



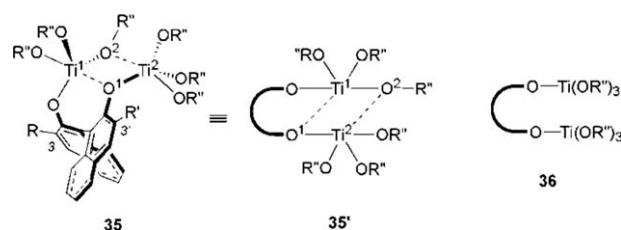
Scheme 33. Enantioselectivity of 3-aryl H₈-BINOLs at 2 mol % and 0.5 mol %.

catalyst loadings but also expands the scope of organometallic reagents to be used as the mixed reagent with Ti(O*i*Pr)₄. Although organotitanium triisopropoxide is likely a reacting species generated in such mixed titanium reagents, mechanistic study was hampered by the complexity of the reaction system. The arylation reaction of aldehydes with ArTi(O*i*Pr)₃ provided us with a relatively simple system to study the mechanism of the titanium catalysis.

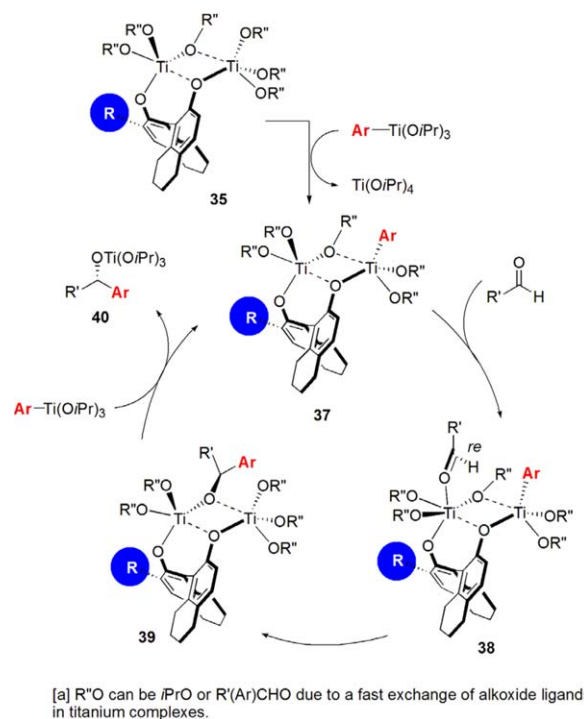
The enantioselectivity of H₈-BINOL and the 3-aryl derivatives in the reaction of 1-naphthaldehyde with PhTi(O*i*Pr)₃ is summarized in Scheme 33.^[48] The parent H₈-BINOL **3** exhibits moderate selectivity at 2 mol % loading. On the other hand, 3-aryl derivatives all show high selectivity (>97% *ee*). At 0.5 mol % catalyst loading, high selectivity (>95% *ee*) is maintained for the DPP and DTBP derivatives but not for the 3,5-dimethylphenyl or phenyl derivatives. Under similar conditions, PhTi(O*i*Pr)₃ reacts with the aldehyde in the absence of the ligands to give the racemic product in 84% yield. The results indicate that rapid non-catalyzed addition of PhTi(O*i*Pr)₃ is responsible for the observed dependence of the enantioselectivity on the catalyst amount. Thus, at 2 mol % loading, the 4-aryl derivatives equally exhibit their inherent high selectivity (>94% *ee*) with a negligible intervention of the background non-catalyzed racemic reaction. At the lower loading, the background racemic process degrades the selectivity by competing with the catalytic enantioselective process. These results strongly suggest that the high turnover frequency of the resulting titanium catalysts is responsible for the excellent performance of the H₈-BINOL derivatives bearing a sterically demanding substituent at the 3-position.

6.2. Origin of High Catalytic Activity

The structures of titanium complexes derived from BINOL derivatives and Ti(O*i*Pr)₄ have been extensively studied in the solid state by X-ray crystallography and in solution by NMR.^[11b,51] With a 1:2 ligand/Ti ratio, bis-titanium complex **35** has been characterized for BINOL^[51a] and H₈-BINOL



Scheme 34. Structure of bis-titanium complexes **35** (R'' = *i*Pr).



Scheme 35. Plausible catalytic cycle for titanium-catalyzed arylation.^[a]

(Scheme 34).^[11b] For the H₈-BINOL-derived complex (**35**; R, R' = H), bond lengths Ti(1)–O(1) and Ti(2)–O(2) (2.11–2.14 Å) are longer than Ti(1)–O(2) (1.92–1.93 Å) and Ti(2)–O(1) (1.98–1.99 Å), implying that the titanium complex can be regarded as the intramolecular aggregate of the two ArO–Ti(O*i*Pr)₃ moieties (**35'**). Inspection of the molecular structure shows that the substitution of a sterically demanding aryl group is allowable only at the 3-position of the 1,1'-biaryl-2,2'-diolate ligand (i.e., R = Ar, R' = H). Indeed, we were successful in the isolation of the DPP-H₈-BINOL derivative (**35**; R = 3,5-diphenylphenyl, R' = H) and confirmed the structure by X-ray analysis.^[52] In comparison with the parent H₈-BINOL derivative, the bridging Ti(1)–O(1) and Ti(2)–O(2) bond distances of the DPP-H₈-BINOL derivative are longer by 0.02 and 0.05 Å, respectively, implying weakening of intramolecular aggregation by the steric constraint around the 3,5-diphenylphenyl group.

It is known that Ti(IV) alkoxides and aryloxides undergo rapid ligand exchange in solution.^[53] According to the previous NMR studies, the *i*PrO groups of bis-titanium complex **35** (R,R' = H) are exchanged both through an intramolecular process and through an intermolecular process with external Ti(O*i*Pr)₄.^[51a] The rate of the processes has been evaluated by VT-NMR experiments.^[51a,52] We carried out the VT-NMR measurement of the DPP-H₈-BINOL-derived complex and observed rate enhancement both for the intra- and intermolecular process in comparison with the H₈-BINOL and BINOL derivatives.^[52] The characteristic behavior of the DPP-H₈-BINOL-derived complex is rationalized by the decrease in the degree of the intramolecular aggregation, facilitating the formation of tetracoordinate bis-titanium intermediate **36**, which might also contribute to the intermolecular *i*PrO group exchange through an associative mechanism.

A plausible catalytic cycle for the titanium-catalyzed arylation is outlined in Scheme 35.^[54] First, the aryl group on ArTi(O*i*Pr)₃ is transferred to **35** to generate complex **37**. Recently, the isolation and X-ray crystal analysis of the (S)-BINOL derivative (**37**; Ar = Ph, R = H) has been reported by Gau et al.^[55] Complex **37** is then coordinated by an aldehyde to form activated complex **38**, which undergoes facile arylation to give intermediate **39** enantioselectively. In complex **38**, substituent R at the 3-position is located distal to the reaction center, which is consistent with the fact that the substituents are not influential on the inherent enantioselectivity of the 3-aryl H₈-BINOL derivatives. Complex **39** then undergoes aryl group transfer with ArTi(O*i*Pr)₃, giving rise to the formation of product titanium alkoxide **40** with simultaneous regeneration of complex **37**.

The catalyst turnover step (**39** + ArTi(O*i*Pr)₃ → **37** + **40**) is closely relevant to the intermolecular exchange process observed for the bis-titanium **35** with external Ti(O*i*Pr)₄, which is accelerated by the introduction of the sterically demanding aryl group at the 3-position. Therefore, it is probable that the rate of aryl group transfer with ArTi(O*i*Pr)₃ is enhanced as well. Assuming that the catalyst turnover step is rate determining, the remarkable enhancement of the turnover frequency observed for the DPP- and DTBP-H₈-BINOL-derived catalyst systems is rationalized by the acceleration of this step due to the steric destabilization of the intramolecular aggregation for intermediate **39**.

7. Conclusion

It was demonstrated that chiral titanium complexes derived from BINOL derivatives bearing a sterically demanding aryl group at the 3-position exhibit high catalytic activity in the enantioselective alkylation, vinylation, arylation, and heteroarylation of aldehydes. Various organometallic reagents, including Grignard, organolithium, organoboron, organozinc, and

organoaluminum reagents, can be employed as the carbon nucleophile. The enantioselective addition of functionalized alkyl and aryl groups is possible using appropriately functionalized organometallic reagents. The organometallic reagents are used with Ti(O*i*Pr)₄ to generate the organotitanium derivatives as the reacting species. Although the organotitanium species are reactive towards the carbonyl group, the activity of the chiral titanium catalyst is high enough to overwhelm the background racemic reaction. Taking advantage of the high catalyst activity, the reactions are carried out at low catalyst loadings of less than 5 mol %. In some instances, 0.5 mol % of the catalyst is sufficient to obtain high enantioselectivities. Despite low catalyst loadings, the reactions proceed rapidly and are accomplished within several hours.

The present results suggest that, in the catalytic enantioselective addition reaction, the reacting organometallic species is not necessarily non-reactive or less reactive towards a carbonyl group, provided that the catalytic pathway is fast enough to override the background racemic reaction.^[56] To realize a catalyst system with high turnover frequency, rate enhancement is indispensable both for the carbonyl addition step and for the catalyst turnover step of the catalytic cycle. The use of reactive organometallic reagents might be advantageous in accelerating the addition step. Rapid ligand exchange often observed in metal alkoxide and aryloxide complexes can be exploited to accelerate the catalyst turnover step. The survey of new catalyst systems based on metals other than titanium is promising for the future development of catalytic enantioselective carbonyl additions of practical utility.

Acknowledgements

I would like to thank Prof. T. Kusakawa, Dr. R. Kumar, Dr. P. A. Adate, Y. Muramatsu, A. Uenishi, Y. Nakagawa, H. Kawasaki, Y. Kinoshita, S. Kanehira, H. Morimoto, Y. Hayashi, T. Ukon, T. Shono, and D. Itakura for their collaboration. The works presented herein were supported financially by Grants-in-Aid from the Ministry of Education, Culture, Sports, Science and Technology (Kakenhi).

REFERENCES

- [1] M. Beetti, E. Lucchi, *Boll. Sci. Fac. Chim. Ind. Bologna* **1940**, 2–5; *Chem. Abstr.* **1940**, 34, 2354.
- [2] H. L. Cohen, G. F. Wright, *J. Org. Chem.* **1953**, 18, 432–446.
- [3] a) H. Nozaki, T. Aratani, T. Toraya, *Tetrahedron Lett.* **1968**, 9, 4097–4098; b) H. Nozaki, T. Aratani, T. Toraya, R. Noyori, *Tetrahedron* **1971**, 27, 905–913; c) D. Seebach, H. Doerr, B. Bastani, V. Ehrig, *Angew. Chem. Int. Ed. Engl.* **1969**, 8, 982–983; d) D. Seebach, H.-O. Kalinowski, B. Bastani, G. Crass, H. Daum, H. Dorr, N. P. DuPreez, V. Ehrig, W. Langer, C. Niessler, H.-A. Oei, M. Schmidt, *Helv. Chim. Acta* **1977**, 60, 301–325; e) T. Mukaiyama, K. Soai, T. Sato, H. Shimizu, K. Suzuki, *J. Am.*

- Chem. Soc.* **1979**, *101*, 1455–1460; f) J.-P. Mazaleyalat, D. J. Cram, *J. Am. Chem. Soc.* **1981**, *103*, 4585–4586.
- [4] Review on the enantioselective carbonyl addition of Grignard reagents and organolithium reagents: M. R. Luderer, W. F. Bailey, M. R. Luderer, J. D. Fair, R. J. Dancer, M. B. Sommer, *Tetrahedron: Asymmetry* **2009**, *20*, 981–998.
- [5] a) N. Oguni, T. Omi, Y. Yamamoto, A. Nakamura, *Chem. Lett.* **1983**, *12*, 841–842; b) N. Oguni, T. Omi, *Tetrahedron Lett.* **1984**, *25*, 2823–2824.
- [6] a) M. Kitamura, S. Suga, K. Kawai, R. Noyori, *J. Am. Chem. Soc.* **1986**, *108*, 6071–6072; b) R. Noyori, S. Suga, K. Kawai, S. Okuda, M. Kitamura, N. Oguni, M. Hayashi, T. Kaneko, Y. Matsuda, *J. Organomet. Chem.* **1990**, *382*, 19–37; c) M. Kitamura, S. Okada, S. Suga, R. Noyori, *J. Am. Chem. Soc.* **1989**, *111*, 4028–4036; d) M. Yamakawa, R. Noyori, *J. Am. Chem. Soc.* **1995**, *117*, 6327–6335. For reviews, see: e) R. Noyori, M. Kitamura, *Angew. Chem. Int. Ed. Engl.* **1991**, *30*, 49–69; f) R. Noyori, *Asymmetric Catalysis in Organic Synthesis*, Wiley, New York, **1994**, Chapter 5.
- [7] Representative reviews: a) K. Soai, S. Niwa, *Chem. Rev.* **1992**, *92*, 833–856; b) L. Pu, H.-B. Yu, *Chem. Rev.* **2001**, *101*, 757–824; c) L. Pu, *Tetrahedron* **2003**, *59*, 9873–9886; d) S. E. Denmark, J. Fu, *Chem. Rev.* **2003**, *103*, 2763–2794; e) M. Hatano, T. Miyamoto, K. Ishihara, *Curr. Org. Chem.* **2006**, *11*, 127–157; f) F. Schmidt, R. T. Stemmler, J. Rudolph, C. Bolm, *Chem. Soc. Rev.* **2006**, *35*, 454–470; g) B. M. Trost, A. H. Weiss, *Adv. Synth. Catal.* **2009**, *351*, 963–983; h) M. Yus, J. C. Gonzales-Gomez, F. Foubelo, *Chem. Rev.* **2011**, *111*, 7774–7854; i) A. Lumbroso, M. L. Cooke, B. Breit, *Angew. Chem. Int. Ed.* **2013**, *52*, 1890–1932; j) H. Pellissier, *Tetrahedron* **2015**, *71*, 2487–2524.
- [8] J. Shannon, D. Bernier, R. Daniel, S. Woodward, *Chem. Commun.* **2007**, 3945–3947.
- [9] For the preparation of higher molecular weight diorganozinc reagents and their use in the catalytic enantioselective addition, see: a) F. Langer, L. Schwink, A. Devasagayaram, P.-Y. Chavant, P. Knochel, *J. Org. Chem.* **1996**, *61*, 8229–8243; b) C. Lutz, P. Knochel, *J. Org. Chem.* **1997**, *62*, 7895–7898; c) C. Bolm, J. Rudolph, *J. Am. Chem. Soc.* **2002**, *124*, 14850–14851; d) F. Schmidt, J. Rudolph, C. Bolm, *Adv. Synth. Catal.* **2007**, *349*, 703–708; e) J. G. Kim, P. J. Walsh, *Angew. Chem. Int. Ed.* **2006**, *45*, 4175–4178; f) L. Salvi, J. G. Kim, P. J. Walsh, *J. Am. Chem. Soc.* **2009**, *131*, 12483–12493; g) A. Côté, A. B. Charette, *J. Am. Chem. Soc.* **2008**, *130*, 2771–2773; h) A. M. DeBerardinis, M. Turlington, L. Pu, *Org. Lett.* **2008**, *10*, 2709–2712; i) D. Glynn, J. Shannon, S. Woodward, *Chem. Eur. J.* **2010**, *16*, 1053–1060; j) M. Hatano, T. Mizuno, K. Ishihara, *Chem. Commun.* **2010**, 46, 5443–5445; k) M. Hatano, R. Gouzu, T. Mizuno, H. Abe, T. Yamada, K. Ishihara, *Catal. Sci. Technol.* **2011**, *1*, 1149–1158.
- [10] a) M. Yoshioka, T. Kawakita, M. Ohno, *Tetrahedron Lett.* **1989**, *30*, 1657–1660; b) H. Takahashi, T. Kawakita, M. Ohno, *Tetrahedron Lett.* **1989**, *30*, 7095–7098; c) H. Takahashi, T. Kawakita, M. Ohno, M. Yoshioka, S. Kobayashi, *Tetrahedron* **1992**, *48*, 5691–5700.
- [11] a) B. Weber, D. Seebach, *Tetrahedron* **1994**, *50*, 7473–7484; b) J. Balsells, T. J. Davis, P. Carroll, P. J. Walsh, *J. Am. Chem. Soc.* **2002**, *124*, 10336–10348.
- [12] a) B. Schmidt, D. Seebach, *Angew. Chem. Int. Ed. Engl.* **1991**, *30*, 99–101; b) B. Schmidt, D. Seebach, *Angew. Chem. Int. Ed. Engl.* **1991**, *30*, 1321–1323.
- [13] a) M. Mori, T. Nakai, *Tetrahedron Lett.* **1997**, *38*, 6233–6236; b) F.-Y. Zhang, C.-W. Yip, R. Cao, A. S. Chan, *Tetrahedron: Asymmetry* **1997**, *8*, 585–589; c) F.-Y. Zhang, A. S. C. Chan, *Tetrahedron: Asymmetry* **1997**, *8*, 3651–3655; d) Y. Tian, K. S. Chan, *Tetrahedron Lett.* **2000**, *41*, 8813–8816; e) S. J. Lee, A. Hu, W. Lin, *J. Am. Chem. Soc.* **2002**, *124*, 12948–12949.
- [14] a) J. M. Brunel, *Chem. Rev.* **2005**, *105*, 857–897; b) J. M. Brunel, *Chem. Rev.* **2007**, *107*, PR1–PR45.
- [15] See reference [11a].
- [16] M. Sakai, M. Ueda, N. Miyaura, *Angew. Chem. Int. Ed.* **1998**, *37*, 3279–3281.
- [17] a) H.-F. Duan, J.-H. Xie, W.-J. Shi, Q. Zhang, Q.-L. Zhou, *Org. Lett.* **2006**, *8*, 1479–1481; b) D. Tomita, M. Kanai, M. Shibasaki, *Chem. Asian J.* **2006**, *1–2*, 161–166; c) T. Nishimura, H. Kumamoto, M. Nagaosa, T. Hayashi, *Chem. Commun.* **2009**, 5713–5715; d) J. Karthikeyan, M. Jegannathan, C.-H. Cheng, *Chem. Eur. J.* **2010**, *16*, 8989–8992; e) S. Morikawa, K. Michigami, H. Amii, *Org. Lett.* **2010**, *12*, 2520–2523.
- [18] Y. Yamamoto, K. Kurihara, N. Miyaura, *Angew. Chem. Int. Ed.* **2009**, *48*, 4414–4416 b) Y. Yamamoto, T. Shirai, M. Watanabe, K. Kurihara, N. Miyaura, *Molecules* **2011**, *16*, 5020–5034.
- [19] a) T. Harada, K. Kanda, *Org. Lett.* **2006**, *8*, 3817–3819; b) T. Harada, T. Ukon, *Tetrahedron: Asymmetry* **2007**, *18*, 2499–2502.
- [20] a) D. Seebach, B. Weidmann, L. Widler, *Mod. Synth. Methods* **1983**, *3*, 217–353; b) M. T. Reetz, *Organotitanium Reagents in Organic Synthesis*, Springer, Berlin, **1986**; c) M. Ishiguro, N. Ikeda, H. Yamamoto, *J. Org. Chem.* **1982**, *47*, 2225–2257; d) N. Tokunaga, K. Yoshida, T. Hayashi, *Proc. Natl. Acad. Sci. U.S.A.* **2004**, *101*, 5445–2449; e) T. Hayashi, N. Tokunaga, K. Inoue, *Org. Lett.* **2004**, *6*, 305–307.
- [21] a) Y. Muramatsu, T. Harada, *Angew. Chem. Int. Ed.* **2008**, *47*, 1088–1090; b) Y. Muramatsu, S. Kanehira, M. Tanigawa, Y. Miyawaki, T. Harada, *Bull. Chem. Soc. Jpn.* **2010**, *83*, 19–32.
- [22] Da and co-workers reported the enantioselective alkylation of aldehydes catalyzed by a BINOL-derived titanium complex (15–20 mol %) using Grignard reagents in the presence of bis[2-(*N,N'*-dimethylamino)ethyl] ether, as a chelating reagent, and Ti(OiPr)₄. a) C.-S. Da, J.-R. Wang, X.-G. Yin, X.-Y. Fan, Y. Liu, S.-L. Yu, *Org. Lett.* **2009**, *11*, 5578–5581; b) Y. Liu, C.-S. Da, S.-L. Yu, X.-G. Yin, J.-R. Wang, X.-Y. Fan, W.-P. Li, R. Wang, *J. Org. Chem.* **2010**, *75*, 6869–6878. For their relevant reports on the enantioselective arylation using Grignard reagents, see: c) X.-Y. Fan, Y.-X. Yang, F.-F. Zhuo, S.-L. Yu, X. Li, Q.-P. Guo, Z.-X. Du, C.-S. Da, *Chem. Eur. J.* **2010**, *16*, 7988–7991; d) L. Zhang, L. Sun, Y.-Y. Li, Y. Liu, Y.-X. Yang, R. Yuan, P. Wang, C.-S. Da, *ChemCatChem* **2013**, *5*, 3516–3519.
- [23] Recently, Yus and co-workers reported that, under similar conditions, (*S_a*,*R*)-1,1'-binaphthalene-2- α -phenylmethan-2'-ol (20 mol %) exhibits high enantioselectivity in the alkylation of aldehydes using Grignard reagents. a) E. Fernandez-Mateos, B. Macia, D. J. Ramon, M. Yus, *Eur. J. Org. Chem.* **2011**, 6851–6855; b) E. Fernandez-Mateos, B. Macia, M. Yus, *Adv. Synth. Catal.* **2013**, *355*, 1249–1254. See also: c) L.-S. Zheng, K.

- Jiang, Y. Deng, X.-F. Bai, G. Gao, F.-L. Gu, L.-W. Xu, *Eur. J. Org. Chem.* **2013**, 748–755.
- [24] Y. Muramatsu, T. Harada, *Chem. Eur. J.* **2008**, *14*, 10560–10563.
- [25] Y. Nakagawa, Y. Muramatsu, T. Harada, *Eur. J. Org. Chem.* **2010**, 6535–6538.
- [26] D. Itakura, T. Harada, *Synlett* **2011**, 2875–2879.
- [27] a) V. A. Vu, I. Marek, K. Polborn, P. Knochel, *Angew. Chem. Int. Ed.* **2002**, *41*, 351–352. For a review, see: b) P. Knochel, W. Dohle, N. Gommermann, F. F. Kneisel, F. Kopp, T. Korn, I. Sapountzis, V. A. Vu, *Angew. Chem. Int. Ed.* **2003**, *42*, 4302–4320.
- [28] a) A. Krasovskiy, P. Knochel, *Angew. Chem. Int. Ed.* **2004**, *43*, 3333–3336; b) L. Shi, Y. Chu, P. Knochel, H. Mayr, *J. Org. Chem.* **2009**, *74*, 2760–2764. For a review, see: c) R. L.-Y. Bao, R. Zhao, L. Shi, *Chem. Commun.* **2015**, *51*, 6884–6900.
- [29] P. Knochel, H. Leuser, L.-Z. Gong, S. Perrone, F. F. Kneisel, in *The Chemistry of Organozinc Compounds* (Eds.: Z. Rappoport, I. Marek), Wiley, Chichester, **2006**, pp. 287–393.
- [30] a) M. J. Rozema, C. Eisenberg, H. Luetjens, R. Ostwald, K. Belyk, P. Knochel, *Tetrahedron Lett.* **1993**, *34*, 3115–3118; b) C. Eisenberg, P. Knochel, *J. Org. Chem.* **1994**, *59*, 3760–3761; c) C. Lutz, P. Knochel, *J. Org. Chem.* **1997**, *62*, 7895–7898.
- [31] T. Ukon, T. Harada, *Eur. J. Org. Chem.* **2008**, 4405–4407.
- [32] R. Kumar, H. Kawasaki, T. Harada, *Org. Lett.* **2013**, *15*, 4198–4201.
- [33] The reaction of $(C_6F_5)_3B$ with $Ti(OiPr)_4$ to give $C_6F_5Ti(OiPr)_3$ and $(C_6F_5)_2BOiPr$ has been reported: A. Rodriguez-Delgado, E. Y.-X. Chen, *Inorg. Chim. Acta* **2004**, *357*, 3911–3919.
- [34] K.-H. Wu, H.-M. Gau, *J. Am. Chem. Soc.* **2006**, *128*, 14808–14809.
- [35] R. D. Rieke, P. T. Li, T. P. Burns, S. T. Uhm, *J. Org. Chem.* **1981**, *46*, 4323–4324.
- [36] A. Krasovskiy, V. Malakhov, A. Gavryushin, P. Knochel, *Angew. Chem. Int. Ed.* **2006**, *45*, 6040–6044.
- [37] E. Negishi, X. Zeng, Z. Tan, M. Qian, Q. Hu, Z. Huang, in *Metal-Catalyzed Cross-Coupling Reactions* (Eds.: A. de Meijere, F. Diederich), 2nd edition, Wiley-VCH, Weinheim, **2004**, pp. 815–877.
- [38] For a non-enantioselective reaction, see: A. Metzger, S. Bernhardt, G. Manolikakes, P. Knochel, *Angew. Chem. Int. Ed.* **2010**, *49*, 4665–4668.
- [39] Y. Kinoshita, S. Kanehira, Y. Hayashi, T. Harada, *Chem. Eur. J.* **2013**, *19*, 3311–3314.
- [40] a) L. Jin, C. Liu, J. Liu, F. Hu, Y. Lan, A. S. Batsanov, J. A. K. Howard, T. B. Marder, A. Lei, *J. Am. Chem. Soc.* **2009**, *131*, 16656–16657; b) E. Hevia, J. Z. Chua, P. García-Álvarez, A. R. Kennedy, M. D. McCall, *Proc. Nat. Acad. Sci. U.S.A.* **2010**, *107*, 5294–5299; c) D. R. Armstrong, W. Clegg, P. García-Álvarez, A. R. Kennedy, M. D. McCall, L. Russo, E. Hevia, *Chem. Eur. J.* **2011**, *17*, 8333–8341.
- [41] T. Shono, T. Harada, *Org. Lett.* **2010**, *12*, 5270–5273.
- [42] F. Gao, A. H. Hoveyda, *J. Am. Chem. Soc.* **2010**, *132*, 10961–10963.
- [43] R. Kumar, H. Kawasaki, T. Harada, *Chem. Eur. J.* **2013**, *19*, 17707–17710.
- [44] For titanium-catalyzed enantioselective addition of alkyl- and arylaluminum reagents, see: a) A. S. C. Chan, F.-Y. Zhang, C.-W. Yip, *J. Am. Chem. Soc.* **1997**, *119*, 4080–4081; b) K. Biswas, O. Prieto, P. J. Goldsmith, S. Woodward, *Angew. Chem. Int. Ed.* **2005**, *44*, 2232–2234; c) reference [34]; d) S. Zhou, K.-H. Wu, C.-A. Chen, H.-M. Gau, *J. Org. Chem.* **2009**, *74*, 3500–3505.
- [45] a) D. E. Van Horn, E. Negishi, *J. Am. Chem. Soc.* **1978**, *100*, 2252–2254; b) E. Negishi, D. Y. Kondakov, D. Choueiry, K. Kasai, T. Takahashi, *J. Am. Chem. Soc.* **1996**, *118*, 9577–9588; c) P. Wipf, S. Lim, *Angew. Chem. Int. Ed. Engl.* **1993**, *32*, 1068–1071; d) B. H. Lipshutz, T. Butler, A. Lower, J. Servesko, *Org. Lett.* **2007**, *9*, 3737–3740.
- [46] H. Morimoto, T. Harada, *Eur. J. Org. Chem.* **2015**, 7378–7383.
- [47] For aryltitanium reagents, see: a) K.-H. Wu, S. Zhou, C.-A. Chen, M.-C. Yang, R.-T. Chiang, C.-R. Chen, H.-M. Gau, *Chem. Commun.* **2011**, *47*, 11668–11670. For alkyltitanium reagents, see: b) Q. Li, H.-M. Gau, *Chirality* **2011**, *23*, 929–939.
- [48] A. Uenishi, Y. Nakagawa, H. Osumi, T. Harada, *Chem. Eur. J.* **2013**, *19*, 4896–4905.
- [49] For a relevant reaction using aryllithium reagents generated from aryl bromides, see: Y.-X. Yang, Y. Liu, L. Zhang, Y.-E. Jia, P. Wang, F.-F. Zhuo, X.-T. An, C.-S. Da, *J. Org. Chem.* **2014**, *79*, 10696–10702.
- [50] For catalytic enantioselective heteroarylation of aldehydes, see: a) reference [9f]; b) X. Liu, L. Qiu, L. Hong, W. Yan, R. Wang, *Tetrahedron: Asymmetry* **2009**, *20*, 616–620; c) A. M. DeBerardinis, M. Turlington, J. Ko, L. Sole, L. Pu, *J. Org. Chem.* **2010**, *75*, 2836–2850.
- [51] a) T. J. Boyle, D. L. Barnes, J. A. Heppert, L. Morales, F. Takusagawa, *Organometallics* **1992**, *11*, 1112–1126; b) J. P. Corden, W. Errington, P. Moore, M. G. Partridge, M. G. H. Wallbridge, *Dalton Trans.* **2004**, 1846–1851; c) G. Pescitelli, L. Di Bari, P. Salvadori, *Organometallics* **2004**, *23*, 4223–4229; d) H. Z. Tang, P. D. Boyle, B. M. Novak, *J. Am. Chem. Soc.* **2005**, *127*, 2136–2142; e) K. Mikami, Y. Matsumoto, L. Xu, *Inorg. Chim. Acta* **2006**, *359*, 4159–4446.
- [52] Y. Hayashi, N. Yamamura, T. Kusukawa, T. Harada, unpublished results.
- [53] a) D. C. Bradley, R. C. Mehrotra, D. P. Gaur, *Metal Alkoxides*, Academic Press, New York, **1978**; b) M. G. Finn, K. B. Sharpless, *J. Am. Chem. Soc.* **1991**, *113*, 113–126; c) Y.-T. Wu, Y.-C. Ho, C.-C. Lin, H.-M. Gau, *Inorg. Chem.* **1996**, *35*, 5948–5952; d) Y. Pérez, I. del Hierro, M. Fajardo, A. Otero, *J. Organomet. Chem.* **2003**, *679*, 220–228.
- [54] For mechanistic studies on relevant chiral titanium-catalyzed reactions, see: a) D. Seebach, D. A. Plattner, A. K. Beck, Y. M. Wang, D. Hunziker, *Helv. Chim. Acta* **1992**, *75*, 2171–2209; b) Y. N. Ito, X. Ariza, A. K. Beck, A. Bohác, C. Ganter, R. E. Gawley, F. N. M. Kühnle, J. Tuleja, Y. M. Wang, D. Seebach, *Helv. Chim. Acta* **1994**, *77*, 2071–2110; c) K.-H. Wu, H.-M. Gau, *Organometallics* **2004**, *23*, 580–588; d) reference [11b].
- [55] K.-H. Wu, Y.-Y. Kuo, C.-A. Chen, Y.-L. Huang, H.-G. Gau, *Adv. Synth. Catal.* **2013**, *355*, 1001–1008.
- [56] Very recently, two kinds of catalytic enantioselective carbonyl addition reaction with reactive organometallic reagents have

been reported. Copper-catalyzed addition of Grignard reagents to ketones: a) A. V. R. Madduri, S. R. Harutyunyan, A. J. Minnaard, *Angew. Chem. Int. Ed.* **2012**, *51*, 3164–3167; b) J. Rong, R. Oost, A. Desmarchelier, A. J. Minnaard, S. R. Harutyunyan, *Angew. Chem. Int. Ed.* **2015**, *54*, 3038–3042. Chiral lithium complex catalyzed addition of alkynyllithiums to ketones: c) K. Tanaka, K. Kukita, T. Ichibakase, S. Kotani,

M. Nakajima, *Chem. Commun.* **2011**, *47*, 5614–5616; d) S. Kotani, K. Kukita, K. Tanaka, T. Ichibakase, M. Nakajima, *J. Org. Chem.* **2014**, *79*, 4817–4825.

Received: October 31, 2015

Published online: April 20, 2016